Advanced Programming Practice · 21CSC203P

Unit 4 · Session 3 · SLO 1 — Solved Worksheet

Name: DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

Unit â€“ 4 : Session â€“ 3 : SLO â€“ 1 | Custom Higher-Order Functions

■ I. Fill in the Blanks

1. lambda â€” a lambda function in Python is a short anonymous function.
2. function â€” a higher-order function can take another function as an argument (or return one).

■ II. Multiple Choice

1. list(filter(lambda x: x % 2 == 0, [1, 2, 3, 4])) â†’ [2, 4] â€” filter keeps only the even numbers.

■ III. Example â€” Custom Higher-Order Function

• def apply_twice(func, value): return func(func(value))

• apply_twice(lambda x: x * 3, 2) # â†’ 18

• Higher-order functions treat functions as first-class values: they can be passed, returned and stored like any other object.